

Design and Implementation of a Flight Controller for a Quadrotor Unmanned Aerial Vehicle

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) require reliable flight controllers, yet existing systems remain closed-source and costly, while open-source alternatives often involve complex codebases. This paper presents the design of an open-source, low-cost flight controller for a quadrotor UAV based on the ESP32 microcontroller. The system integrates an Inertial Measurement Unit (IMU), GPS module, barometric sensor, and wireless telemetry on a custom Printed Circuit Board (PCB) suitable for educational and experimental applications. A dual-loop Proportional–Integral–Derivative (PID) controller is implemented for roll, pitch, and yaw stabilization, with a Complementary filter fusing gyroscope and accelerometer data to reduce noise and drift in attitude estimation. Stability analysis of the linearized plant model yields a damping ratio (ζ) = 0.71 and phase margin (PM) = 81°, validating the controller design. Experimental results demonstrate stable hover performance under real-time conditions. The proposed system offers transparency, reduced cost, and a scalable foundation for future autonomous navigation enhancements.

KEYWORDS : *Quadrotor UAV, Flight controller design, ESP32 Microcontroller, PID control, Complementary filter.*

INTRODUCTION

Recent Advancements in field of UAVs have led to increase in demand of the efficient flight controller. However, existing solutions face challenges such as high cost, closed-source proprietary ecosystems, and complexity that is often excessive for simple applications. The existing flight controllers are closed source and have high cost to procure for educational purpose. Moreover, the open-source platforms come with complex codebase. To address these issues, this paper presents a Cost Effective, Open Source, Open Hardware Architecture with flexible integration. The proposed approach aims designing and implementing a custom flight controller board with integrated sensors and firmware, adaptable for various UAV applications which can serve as an educational open platform for students learning UAV control systems. Its architecture and transparent implementation can make it suitable for academic use, allowing students to understand and experiment with flight control algorithms and sensor integration. Flight Controller acts as the drone's brain processing pilot inputs and sensor data for stable and precise flight. To solve the problem an ESP32-based

system is developed that provides integrated Wi-Fi, sufficient storage, and multiple serial ports in a single, affordable unit and is implemented on a custom designed PCB. To enable reliable navigation and state estimation, the system integrates multiple sensors, including a Motion Processing Unit (MPU6050), a Global Positioning System (GPS) module, barometer, etc. This hardware configuration provides a compact and cost-effective platform suitable for UAV control and experimentation. The design phase involves defining the system architecture, selecting appropriate hardware and software components, and developing the logical flow and control strategy of the system. It ensures that the overall structure of the system meets functional and performance requirements before actual development. The implementation phase involves the practical prototype and PCB circuit of the designed system. This includes hardware assembly and interfacing, software development, integration of various modules, and testing of the system under real-time conditions. The results obtained from the implementation demonstrate that the proposed system operates reliably and meets the intended objectives.

LITERATURE REVIEW

The research demonstrated that a cascade PID controller combined with sensor filtering provides stable and effective flight control for quadrotor UAVs. Previous work explored advanced control strategies to improve altitude response while emphasizing low-complexity and low-power flight controller implementations suitable for embedded systems. The present work adopts a dual-loop cascade PID control structure with inner and outer loops for roll, pitch, and yaw, enabling stable flight performance. Some researchers further suggested the application of Model Predictive Control (MPC) for enhanced trajectory tracking and stabilization in quadrotor UAVs. In addition, research on navigation accuracy emphasized minimizing IMU sensor noise, bias, and drift using Kalman filtering and Zero Velocity Compensation (ZVC) techniques. The Kalman Filter was shown to effectively reduce sensor drift, achieving approximately 95% angular accuracy and 90% translational accuracy. Building on this foundation, the present work evaluates both Kalman filtering and Complementary filtering for attitude estimation. While both approaches demonstrated comparable accuracy, the Complementary filter was ultimately selected for its lower computational complexity and suitability for real-time embedded systems. Further studies proposed real-time embedded implementation to enhance navigation accuracy and computational efficiency [1]-[4]. Several studies focused on stabilizing and controlling quadrotor attitude using an optimized PID controller and compared its performance with alternative nonlinear control methods such as the back-stepping controller. These studies highlighted the fundamental instability of quadrotor attitude dynamics and the necessity of robust feedback control mechanisms. Results showed that an optimized PID controller offers faster response, improved stability, and better disturbance rejection compared to the back-stepping controller [5]. Moreover, sensor fusion approaches using complementary filters were shown to provide reliable attitude estimation, significantly improving PID-based stabilization performance in quadrotor systems [6]. Recent studies have highlighted the growing use of UAVs in applications such as precision agriculture, infrastructure inspection, and environmental monitoring due to their efficiency and data collection capabilities. Research on real-time UAV control emphasized low-latency communication and robust control architectures for stable flight performance. Multi-UAV ground control stations were shown to enable effective coordination

and mission management in complex environments [7]-[9]. Studies on embedded real-time operating systems revealed that Real Time Operating System (RTOS) selection strongly influences performance and energy consumption in UAV flight controllers [10]. Fault-tolerant and optimization-based control approaches, including multi-rate control and Particle Swarm Optimization (PSO) tuned PID controllers, demonstrated improved stability, robustness, and response under disturbances and actuator faults [11], [12]. In addition, actuator fault detection using inertial sensors enhanced UAV safety, while hardware selection surveys provided essential guidelines for designing efficient and reliable UAV platforms [13], [14]. Experimental evaluation showed that both the 1D Kalman filter and the Complementary filter achieved comparable attitude estimation accuracy, maintaining stability within $\pm 0.5^\circ$ while effectively mitigating gyroscope drift and accelerometer noise. Given this equivalent performance, the Complementary filter was selected due to its significantly lower computational complexity, simpler tuning, smaller memory footprint, and better suitability for reliable real-time implementation on resource-constrained flight controllers [15].

METHODOLOGY

System Design

Overall System Architecture

The overall system architecture of the proposed quadrotor flight controller is illustrated in Figure 2, which presents the system-level architecture and functional data flow from onboard sensors through control computation and actuation to telemetry and ground station communication. The proposed quadrotor flight controller follows a modular, layered system architecture that separates sensing, control, actuation, and communication functions to ensure deterministic real-time performance and system reliability. At the lowest level, onboard sensors including an MPU6050, barometric pressure sensor, and GPS module continuously acquire raw motion, altitude, and positional data. These sensor signals are interfaced directly to the ESP32 microcontroller through Inter-Integrated Circuit (I²C) and Universal Asynchronous Receiver Transmitter (UART) communication buses. Within the ESP32, sensor data first undergoes preprocessing and filtering. A complementary filter fuses gyroscope and accelerometer measurements to obtain drift-reduced roll and pitch

estimates, while raw gyroscope data supports yaw rate estimation. The filtered attitude information is then passed to a dual-loop PID control structure, where the outer loop regulates angular position and the inner loop controls angular rates. This control computation is executed on a dedicated processor core to guarantee consistent loop timing. The PID controller outputs are translated into high-resolution Pulse Width Modulation (PWM) signals using the ESP32's dedicated Motor Control PWM (MCPWM) peripheral. These signals are fed to four Electronic Speed Controllers (ESCs), which in turn regulate the speeds of the brushless DC motors, producing the required thrust and torque for quadrotor stabilization and maneuvering. Parallel to the flight-critical control path, a telemetry and diagnostics subsystem operate independently. Flight states, sensor readings, and control outputs are transmitted via the Nordic Radio Frequency (nRF24L01) wireless module to a ground station for real-time monitoring. Additionally, a non-blocking data logging mechanism queues control and sensor data to a secondary core, where it is stored in a circular RAM buffer for post-flight analysis. This architectural separation ensures that telemetry, logging, and communication tasks do not interfere with the real-time stability of the flight control loop.

Hardware Selection

The ESP32 microcontroller was selected over Arduino and STM32 platforms for its dual-core architecture, which enables physical workload separation, Core 1 dedicated to the 250 Hz flight control loop while Core 0 handles telemetry and logging eliminating timing inconsistencies inherent in single-core systems that require context switching between concurrent tasks. With 520 KB RAM (versus Arduino's 2 KB or STM32 Blue Pill's 20 KB), the ESP32 provides adequate memory for floating-point PID control and RTOS operation. Its dedicated MCPWM peripheral generates jitter-free PWM signals independently of the CPU, ensuring precise motor control. Integrated Wi-Fi and Bluetooth reduce hardware complexity compared to STM32 solutions requiring external wireless modules. For positioning, the u-blox NEO-M8N GPS module provides multi-constellation support for faster 3D lock acquisition, transmitting coordinates and ground speed to the ground station for situational awareness. The system implements dual-link communication architecture, separating the critical control path (pilot input via dedicated transmitter) from the telemetry path nRF24L01, ensuring pilot control is retained even if the telemetry link fails.

Black Box

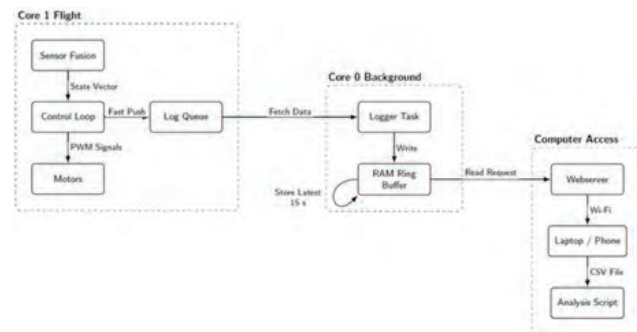


Fig. 1: System architecture and data flow of the black box logging and analysis framework

A custom black box system records time-synchronized flight data (sensor measurements, attitude estimates, PID outputs, actuator commands) for post-flight analysis and PID tuning. To address flash memory latency (10–20 ms) that would disrupt the 250 Hz control loop, the dual-core ESP32 architecture is leveraged as shown in Figure 1. Core 1 executes flight control operations and pushes the complete control state into a FreeRTOS queue using non-blocking operations. Core 0 runs a low-priority logger task that writes queued data into a circular RAM buffer, automatically overwriting older samples when capacity is reached. Post-mission, an embedded HTTP server serializes the buffer to CSV format and transmits it wirelessly via Wi-Fi for analysis. Using volatile RAM ensures minimal write latency and complete isolation between flight-critical and diagnostic subsystems.

Control System Design

System Block Diagram

The flight controller system, illustrated in Figure 3, is built around the ESP32 microcontroller, which acts as the central processing unit of the quadcopter. The UAV platform consists of four brushless DC motors mounted on an F450 frame, controlled by Electronic Speed Controllers (ESC1 to ESC4). An MPU6050 IMU comprising a 3-axis accelerometer and 3-axis gyroscope measures the UAV's orientation and angular rates, while a barometric sensor estimates altitude and a GPS module provides position data. These sensor signals are continuously fed to the ESP32, which processes the data to estimate roll, pitch, and yaw angles. Based on pilot commands received via the transmitter/receiver and sensor inputs, the ESP32 calculates the required control signals and generates PWM signals to drive the ESCs. The system also includes an

nRF24L01 telemetry module for wireless communication with the ground station, enabling real-time monitoring of flight data. Power is supplied using a Lithium Polymer (Li-Po) battery, selected based on required flight time and thrust-to-weight ratio.

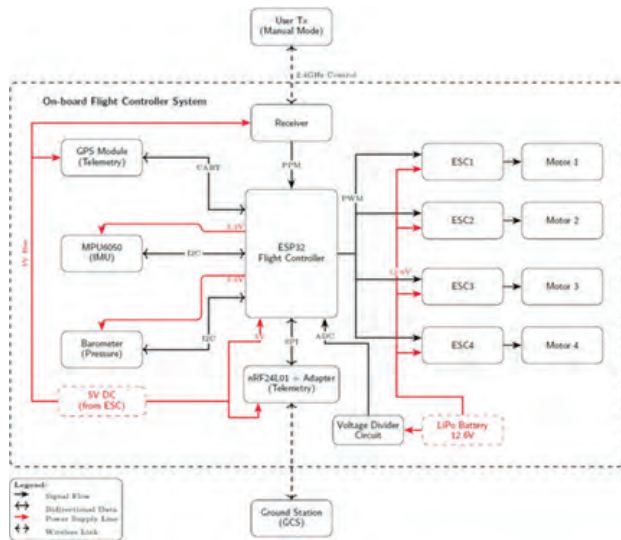


Fig. 2: Block diagram of drone flight controller system

Control System Design

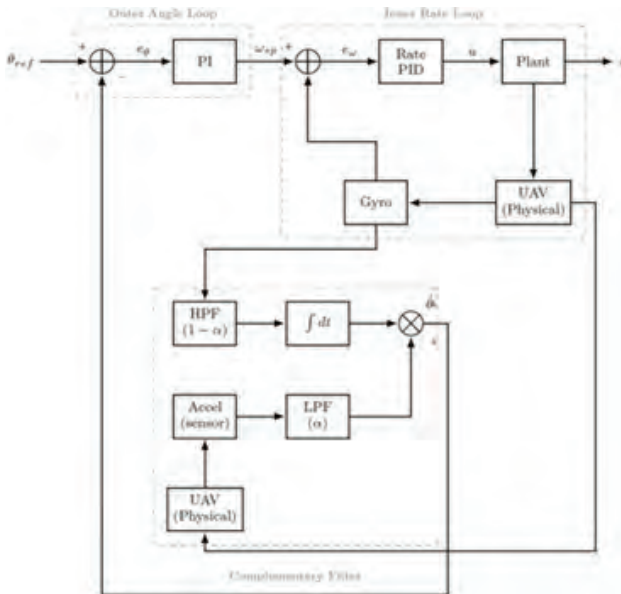


Fig. 3: Cascade control architecture with complementary filter

Signals: θ_{ref} = angle setpoint (deg), e_θ = angle error, ω_{sp} = rate setpoint, e_ω = rate error, u = control output, ω = angular rate, $\hat{\theta}$ = estimated angle.

The stability analysis was conducted on a dual-loop cascade PID control architecture operating at 250 Hz. The system comprises an ESP32 microcontroller, MPU6050 IMU, 1400 KV motors, F450 frame, and 2200 mAh battery, yielding a thrust-to-weight ratio of approximately 3.6:1. The cascade structure consists of two loops: an inner rate loop that stabilizes angular velocity using PID control, and an outer angle loop that generates rate setpoints based on angle error using PI control. Attitude feedback is derived from a complementary filter fusing gyroscope and accelerometer data. Figure 3 illustrates the complete control architecture.

System Modeling

The plant transfer function models ESC dynamics, motor response, and rigid body rotation based on Newton's Euler

$$\text{equation } \tau = I \frac{d\omega}{dt}$$

$$G_p(s) = \frac{K_\tau}{s(\tau_m s + 1)} = \frac{85}{0.030s^2 + s}$$

where $K_\tau = 85 \text{ deg/s}^2$ per unit is torque gain and $\tau_m = 30 \text{ ms}$ is motor time constant.

Controller Design

The inner rate loop implements a PID controller with derivative low-pass filter:

$$G_{c,rate}(s) = K_p + \frac{K_i}{s} + \frac{K_d s}{T_d s + 1}$$

The outer angle loop uses a PI controller:

$$G_{c,angle}(s) = K_{p,angle} + \frac{K_{i,angle}}{s}$$

Attitude feedback is obtained through a complementary filter:

$$\hat{\theta} = (1 - \alpha) \int \omega_{gyro} dt + \alpha \theta_{accel}$$

where $\alpha \approx 0.04$ provides optimal fusion—gyroscope gives accurate short-term measurements while accelerometer corrects long-term drift.

PID Tuning Process

Controller parameters were determined through iterative flight testing using black box log analysis. Table 1 shows the tuning progression.

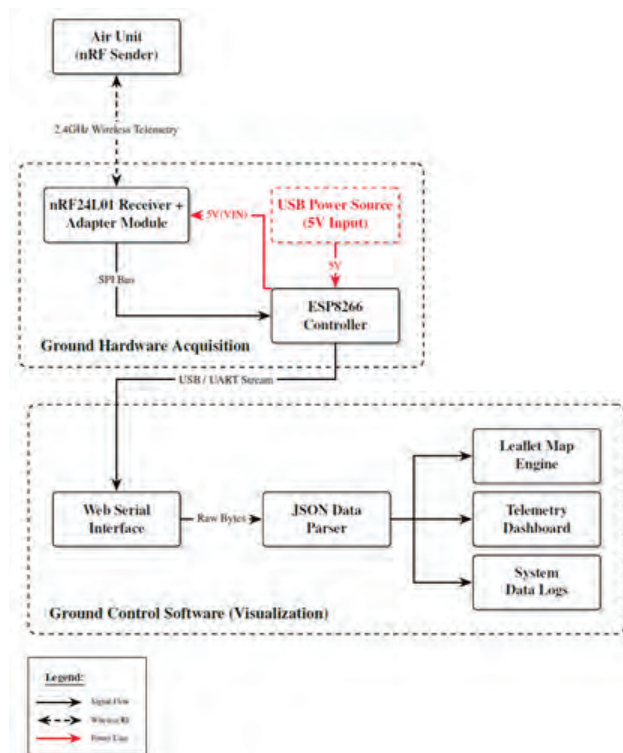
Table 1: PID tuning iterations

Iteration	K_p	K_i	K_d	Outcome
Initial	0.035	0.00	0.00	Sluggish, drifting
Iteration 1	0.10	0.05	0.01	Underpowered response
Iteration 2	0.25	0.10	0.02	Improved, minor oscillation
Iteration 3	0.35	0.15	0.025	Good, slight overshoot
Final	0.40	0.20	0.03	Stable, no oscillation

Table 2: Final Controller parameters

Parameter	Symbol	Value	Unit
Proportional gain	K_p	0.40	-
Integral gain	K_i	0.20	s^{-1}
Derivative gain	K_d	0.03	s
D-term filter constant	T_d	5.5	ms
Control loop frequency	f	250	Hz

Ground Station

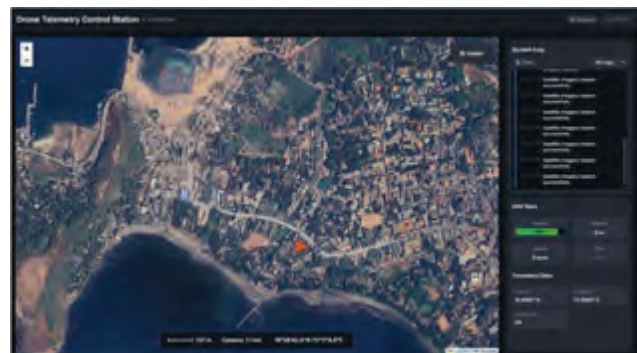
**Fig. 4: Ground station system architecture and data flow**

The ESP8266 microcontroller was selected for the ground station receiver to maintain architectural consistency with the ESP32-based flight controller. Since the ESP32

features integrated Wi-Fi, future implementations can optionally bypass the nRF24L01 radio link and establish direct Wi-Fi communication between the drone and ground station, allowing students or researchers to experiment with Wi-Fi telemetry by simply updating firmware without hardware modifications. The Ground Control Station (GCS) serves as a web-based visualization interface displaying real-time flight parameters including GPS coordinates, altitude, heading, and battery status. Built as a Single Page Application using standard HTML/CSS and JavaScript, the dashboard communicates directly with the ESP8266 over USB port, eliminating the need for a backend server. The interface integrates Leaflet.js library to plot GPS coordinates on an open-source map for real-time flight path visualization. Data is transmitted using JSON protocol, with the ground receiver parsing binary radio packets into human-readable structured text for debugging and analysis.

RESULTS AND DISCUSSION

Figure 5 shows the final output of our drone project, which is a Drone Control Station or Ground Station Interface. It provides a clear and simple way to monitor and control the drone in real time. The main screen displays a map view, where the live position of the drone is shown using GPS data. This helps the user easily track the drone's location and movement during flight. On the right side, important flight information such as battery level, altitude, speed, distance, and GPS status is displayed. Telemetry data like latitude and longitude is also shown for accurate positioning. Overall, this interface makes drone operation easy, safe, and user-friendly by combining live tracking, status monitoring, and control options in one screen.

**Fig. 5: Web based ground control station dashboard interface**

Gyroscope-only estimation drifts over time, while accelerometer readings are noisy during flight. We tested

both Kalman and Complementary filters—both achieved $\pm 0.5^\circ$ accuracy, so we chose the Complementary filter for its simpler implementation. The comparative results (gyroscope, accelerometer, Kalman, and Complementary filter outputs) are shown in Figure 6 in the Results section.

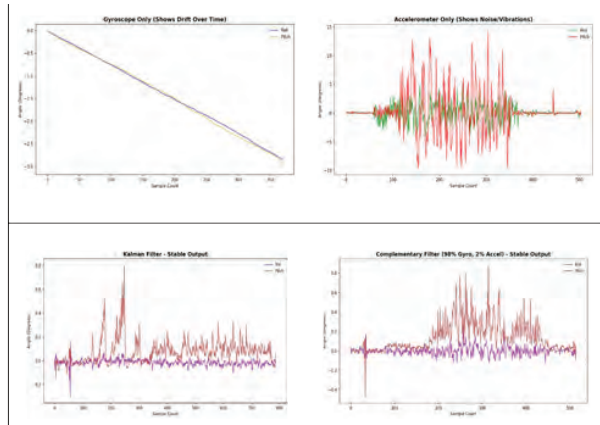


Fig. 6: Sensor fusion comparison: (a) gyroscope drift, (b) accelerometer noise, (c) kalman filter output, (d) complementary filter output



Fig. 7: Assembled quadcopter prototype with custom flight controller

Stability Analysis

Table 3: Stability Margins and Performance

Mode	ζ	PM	GM	Rise Time	Settling Time
Rate	0.71	81°	>20 dB	81 ms	229 ms
Angle	0.96	81°	33.8 dB	424 ms	647 ms

Closed-loop pole analysis confirms the asymptotic stability of the cascade control system. The rate controller achieves a damping ratio of $\zeta = 0.71$, which is close to the optimal value of 0.707, with dominant poles located at $-102.6 \pm 100.4j$, ensuring fast and responsive behavior. The angle controller exhibits an overdamped response with a higher damping ratio of $\zeta = 0.96$ and dominant poles at $-6.7 \pm 2.0j$, resulting in improved stability. A slow pole at $s = -0.1$, introduced by the integral term, ensures zero steady-state error and compensates for gyroscope drift through accelerometer-based attitude feedback. All system poles lie in the left half-plane, confirming overall stability. Frequency-domain analysis shows a phase margin of 81° and gain margin of 33.8 dB, exceeding standard stability requirements ($PM > 45^\circ$, $GM > 6$ dB). The step response demonstrates zero overshoot with a rise time of 424 ms and a settling time of 647 ms. Figure 8 illustrates the stability characteristics of the cascade controller. Comparative analysis indicates that the angle mode provides slower but more stable dynamics than the rate mode. Overall, the developed UAV achieves stable hovering, responsive control, and reliable performance, establishing a solid foundation for future enhancements such as autonomous navigation and advanced control strategies.

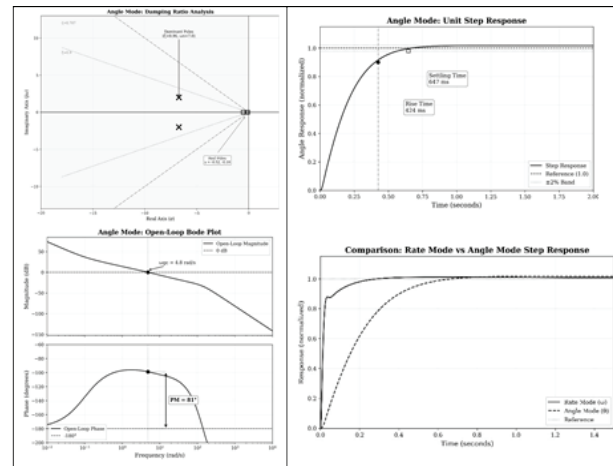


Fig. 8: Stability analysis: S-plane poles, step response, Bode plots, and mode comparison.

CONCLUSION

The paper presented the design and implementation of an open-source, low-cost flight controller for a quadrotor UAV based on the ESP32 microcontroller. The dual-core architecture enables real-time PID control on one core while handling telemetry and data logging on the

other, eliminating timing conflicts inherent in single-core systems. Stability analysis confirmed robust performance with damping ratio $\zeta = 0.96$ and phase margin $PM = 81^\circ$, while experimental testing demonstrated stable hovering and responsive control. Future work includes implementing GPS-based autonomous flight modes (waypoint navigation, return-to-home), exploring extended Kalman filtering for improved attitude estimation during aggressive maneuvers, and field testing under diverse conditions to establish reliability benchmarks for educational UAV applications.

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REFERENCES

1. Waliszewicz, M., Wojtowicz, K., Rochala, Z., & Balestrieri, E. (2020). The Design and Implementation of a Custom Platform for the Experimental Tuning of a Quadcopter Controller. *Sensors (Basel)*.30;20(7):1940.
2. Mirtaba, M., Jeddi, M., Nikoofard, A. (2023). Design and implementation of a low- complexity flight controller for a quadrotor UAV. *Int. J. Dynam. Control* 11, 689–700.
3. Lasmadi, Cahyadi, A., Herdjunto, S. & Hidayat, R. (2017). Inertial Navigation for Quadrotor Using Kalman Filter with Drift Compensation. *International Journal of Electrical and Computer Engineering (IJECE)*. 7. 2596. 10.11591/ijece.v7i5.pp2596- 2604.
4. Kadam, S., & Ruikar, S. (2024). Designing flight controller of quadcopter using STM32 microcontroller. *International Journal of Innovative Research In Technology (IJIRT)*, 11(3), 1163-1168.
5. Bolandi, H., Rezaei, M., Mohsenipour, R., Nemati, H. & Smailzadeh, S. (2013). Attitude Control of a Quadrotor with Optimized PID Controller. *Intelligent Control and Automation*, 4, 335-342.
6. Noordin, A., Basri, A., & Mohamed, Z. (2018). Sensor Fusion for Attitude Estimation and PID Control of Quadrotor UAV. *International Journal of Electrical and Electronic Engineering & Telecommunications*. 7. 183-189. 10.18178/ijeetc.7.4.183-189.
7. Radoglou-Grammatikis, P., Sarigiannidis, P., Lagkas, T. & Moscholios, I. (2020). A compilation of UAV applications for precision agriculture. *Computer Networks* 172:107148.
8. Kangunde, V., Jamisola, R. & Theophilus, E. (2021). A review on drones controlled in real- time. *Int J Dyn Control* 9(4):1832–1846.
9. Poma, A., Sojo, A., Maza, I & Ollero, A. (2025). Ground Control Station for Multi- UAV Systems in Infrastructure Inspection and Environmental Monitoring Applications. *Journal of Intelligent & Robotics Systems*. 111:109
10. Baynes, K., Collins, C., Fiterman, E., Brinda, G., Kohout, P., Christine, S., Zhang, T & Jacob, B. (2003). The Performance and Energy consumption of Embedded Real-Time Operating Systems. *IEEE Transactions on computers*, Vol.52, No.11
11. Yang, T., Bu, K., Chen, G., Xie, X. & Xia, J. (2024). Fault-tolerant multi-rate sampled- data control for quadrotor UAV. *Nonlinear Dyn* 112:12253-12267.
12. Sahrir, N, H. & Basri, A. (2023). PSO-PID Controller for Quadcopter UAV: Index Performance Comparison. *Arabian Journal for Science and Engineering* 48:15241-15255.
13. Zhou, L., Jin, H., Chen, P., Xiong, J., Li, C & Xiong, W. (2025). Actuator fault detection method of quadrotor UAV based on dual channel inertial sensors. *Zhou et al. Discover Applied Sciences* 7:736.
14. Sdoukou, E., Milidonis, A., Efstathiou, K & Voyiatzis, I. (2025). A survey of UAV hardware selection. *Sdoukou et al. Journal of Engineering and Applied Science*. 72:88.
15. Liu, M., Cai, Y., Zhang, L. & Wang, Y. (2021). Attitude Estimation Algorithm of Portable Mobile Robot Based on Complementary Filter. *Micromachines* 2021, 12, 1373.